

COURSE SYLLABUS

CSC15012 – Applications of Natural Language Processing in Industry

1. GENERAL INFORMATION

Course name:	Applications of Natural Language Processing in Industry
Course name (in Vietnamese):	Ứng dụng xử lý ngôn ngữ tự nhiên trong doanh nghiệp
Course ID:	CSC15012
Knowledge block:	Fundamental and specialized knowledge
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	
Instructors:	Nguyen Hong Buu Long

2. COURSE DESCRIPTION

This course introduces the principles and industrial applications of Natural Language Processing (NLP), focusing on the design, deployment, and management of production-ready NLP systems. It covers the NLP lifecycle, including development infrastructure, data management, model optimization, deployment, and continual learning.

The course emphasizes practical industry considerations such as scalability, privacy, robustness, ethics, and project management. Through real-world case studies and demo applications, students gain insight into end-to-end NLP workflows and the challenges of deploying NLP solutions at scale.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Develop a solid understanding of industrial Natural Language Processing (NLP) systems, including the end-to-end lifecycle from data acquisition and model development to deployment and maintenance in real-world environments.	LO1, LO2
G2	Understand modern NLP development infrastructure, tooling, and data management practices, and analyze their role in building scalable, reliable, and efficient NLP applications for industry use.	LO6, LO7
G3	Apply model optimization, deployment, and continual learning techniques to practical NLP problems, with an emphasis on performance evaluation, robustness, and real-world constraints.	LO6, LO7
G4	Evaluate NLP systems using appropriate metrics and methodologies, and critically assess trade-offs related to accuracy, cost, scalability, privacy, and ethical considerations in industrial settings.	LO3, LO4, LO5, LO7
G5	Strengthen professional skills required for industrial NLP projects, including teamwork, project planning, communication with stakeholders, and analysis of real-world case studies and demo applications.	LO12

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Know and understand fundamental terminology, concepts, and scope of Natural Language Processing and its industrial applications.	I

G1.2	Know and understand NLP development infrastructure, tools, and data management practices used in industry.	I
G2.1	Understand and explain NLP models, architectures, and optimization techniques for industrial text processing tasks.	T
G2.2	Apply appropriate NLP models and evaluation metrics to solve real-world language processing problems.	U
G3.1	Understand and analyze deployment strategies for NLP systems, including scalability, monitoring, and continual learning.	T
G3.2	Apply deployment, maintenance, and continual learning techniques in industrial NLP case studies and demo applications.	U
G4.1	Identify issues related to data privacy, model robustness, and ethical considerations in NLP applications.	T
G4.2	Evaluate industrial NLP systems with respect to performance, privacy, robustness, and ethical impact.	U
G5.1	Develop professional skills including teamwork, project planning, and technical communication in NLP projects.	U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduction: - Overview of Natural Language Processing (NLP) and its evolution; - Key NLP tasks (classification, extraction, generation, retrieval, summarization);	G1.1	Lecturing Q&A, Group discussion	

	- Role of NLP in modern industries (tech, healthcare, finance, manufacturing, media); - NLP pipeline: data → model → evaluation → deployment; - Industry vs academic NLP: constraints, scale, and business value.			
2	Development Infrastructure & Tooling: - NLP development environments (local, cloud, notebooks, IDEs); - Frameworks and libraries (tokenizers, ML/DL frameworks, LLM APIs); - Version control and experiment tracking; - Compute infrastructure: CPUs, GPUs, TPUs, cloud services; - MLOps for NLP workflows.	G1.2	Lecturing Q&A, Group discussion	
3	Data Management: - Text data sources: structured, semi-structured, unstructured; - Data collection, cleaning, normalization, and preprocessing; - Labeling strategies and annotation workflows; - Dataset versioning and lifecycle management; - Handling data imbalance and noise.	G1.2, G2.1	Lecturing Q&A, Group discussion	
4	Model & Parameter Optimization: - Model selection: classical NLP vs	G2.1, G2.2	Lecturing Q&A, Group	

	deep learning vs transformers; - Hyperparameter tuning strategies; - Fine-tuning pre-trained language models; - Evaluation metrics for NLP tasks; - Cost-performance trade-offs in industry.		discussion	
5	Deployment: - Packaging and serving NLP models (APIs, microservices); - Real-time vs batch inference; - Scalability, latency, and reliability considerations; - Monitoring model performance in production; - Handling model updates and rollback strategies.	G3.1	Lecturing Q&A, Group discussion	
6	Agentic AI: - Concept of agentic AI and autonomous NLP systems; - NLP agents for reasoning, planning, and tool usage; - Multi-agent systems and orchestration; - Use cases: chatbots, assistants, workflow automation; - Risks and control mechanisms for agentic systems.	G2.1, G3.1	Lecturing Q&A, Group discussion	
7	Continual Learning:	G3.1, G3.2	Lecturing	

	- Concept drift and data drift in NLP systems; - Online and incremental learning approaches; - Retraining strategies and feedback loops; - Human-in-the-loop learning; - Challenges of catastrophic forgetting.		Q&A, Group discussion	
8	Data Privacy & Model Robustness: - Privacy issues in text data (PII, sensitive information); - Data anonymization and masking techniques; - Secure model training and deployment; - Adversarial attacks on NLP systems; - Robustness testing and stress evaluation.	G4.1	Lecturing Q&A, Group discussion	
9	Teams and Project Management: - Roles in industrial NLP projects (data scientists, engineers, annotators); - Agile and MLOps-driven project workflows; - Requirement gathering and problem definition; - Collaboration between technical and business teams; - Managing timelines, risks, and resources.	G5.1	Lecturing Q&A, Group discussion	

10	Ethics: - Bias and fairness in NLP models; - Explainability and transparency; - Responsible AI principles; - Legal and regulatory considerations; - Social impact of large-scale NLP systems;	G4.1, G4.2	Lecturing Q&A, Group discussion	
11	Case Study: Data Annotation & Demo Applications: - End-to-end NLP project walkthrough; - Data annotation process and quality control; - Model training and evaluation on real data; - Demo applications (chatbots, classifiers, summarizers); - Lessons learned and best practices.	G2.2, G3.2, G4.2, G5.1	Lecturing Q&A, Group discussion	

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Lab 1: Industrial NLP pipeline exploration: - Text data collection from open or industrial-style sources; - Data cleaning, preprocessing, and normalization; - Dataset organization and version control;	G1.2, G2.1	Group discussion	LW#1

	- Basic exploratory analysis of text data.			
2	Lab 2: Model Training, Optimization, and Evaluation: - Fine-tuning a pre-trained NLP model; - Hyperparameter optimization; - Model evaluation using accuracy, precision, recall, and F1-score; - Analysis of performance trade-offs.	G2.1, G2.2, G4.2	Group discussion	LW#2
3	Lab 3: Deployment, Ethics, and Case Study Analysis: - Basic model deployment using APIs or services; - Monitoring and error analysis; - Identification of privacy and ethical risks; - Team-based discussion and reporting.	G3.2, G4.1, G4.2, G5.1	Group discussion	LW#3

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments			30%
A11	Lab work: LW1–LW3	Lab 1: Industrial NLP pipeline exploration Lab 2: Model Training,	G1.2, G2.1, G2.1, G2.2, G4.2, G3.2,	30%

		Optimization, and Evaluation Lab 3: Deployment, Ethics, and Case Study Analysis	G4.1, G4.2, G5.1	
A2	Projects			70%
A21	Seminar Presentation	Students will present their group project, demonstrating understanding of the problem domain, applied NLP methods, system design choices, and analysis of results.	G1.1, G1.2, G2.1, G4.1, G5.1	30%
A22	Final Evaluation	Assessment of the final project deliverables, including technical implementation, analytical depth, correctness, innovation, and alignment with industrial practices.	G2.2, G3.2, G4.2, G5.1	40%

7. RESOURCES

Textbooks

- Daniel Jurafsky, James H. Martin. “Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition”, 2025 (3rd ed. Draft).
- Hobson Lane and Maria Dyschel. “Natural Language Processing in Action”, January 2025, ISBN 9781617299445, 688 pages.
- Husn Ara, “Mastering Natural Language Processing: Beginner to Advanced Level Guide”, October 18, 2024, ISBN 979-8343611366, 476 pages.
- Antonio Gullí, Agentic Design Patterns: A Hands-On Guide to Building Intelligent Systems, Springer Cham, 427 pages, 30 October 2025.

- Andrew Davis, “Natural Language Processing with Artificial Intelligence: Building Smarter Applications Through Language Understanding”, October 8, 2024, 140 pages.
- Chip Huyen, AI Engineering, 1st Edition, 534 pages, O'Reilly Media, December, 2024.
- Chip Huyen, Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications, 1st Edition, 386 pages, O'Reilly Media, June 21, 2022.
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Others

- Google DeepMind, How to Scale Your Model: A Systems View of LLMs on TPUs, Feb. 4, 2025. Link: <https://jax-ml.github.io/scaling-book/>

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.